

PANDAS NULL HANDLING - QUICK NOTES

1. CHECK NULLS

```
df.isnull()
df.isnull().sum()
df['col'].isnull()
```

2. FILTER ROWS

```
df[df['marks'].isnull()]
df[df['marks'].notnull()]
df[df.notnull().all(axis=1)]
df[df.notnull().any(axis=1)]
```

3. FILL NULL VALUES

```
df.fillna(0)
df['marks'].fillna(50)
df['marks'].fillna(df['marks'].mean())
df['city'].fillna('Unknown')
```

4. DROP NULL VALUES

```
df.dropna()
df.dropna(subset=['marks'])
df.dropna(axis=1)
```

5. THRESHOLD

df.dropna(thresh=2) → keep rows with at least 2 values

6. FILL METHODS

```
df.fillna(method='ffill')
df.fillna(method='bfill')
```

7. DATA TYPES

NaN = float → column becomes float
df['marks'] = df['marks'].fillna(0).astype(int)

8. COPY

`df2 = df.copy()` (safe copy)

FINAL MEMORY

FIND → `isnull()`

FIX → `fillna()`

REMOVE → `dropna()`